

Homework Assignment 12

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5B.1.

(a) Let λ denote Lebesgue measure on $[0, 1]$. Show that

$$\int_{[0,1]} \int_{[0,1]} \frac{x^2 - y^2}{(x^2 + y^2)^2} d\lambda(y) d\lambda(x) = \frac{\pi}{4}$$

and

$$\int_{[0,1]} \int_{[0,1]} \frac{x^2 - y^2}{(x^2 + y^2)^2} d\lambda(x) d\lambda(y) = -\frac{\pi}{4}.$$

(b) Explain why (a) violates neither Tonelli's Theorem nor Fubini's Theorem.

Proof. Let $f = (x^2 - y^2)/(x^2 + y^2)^2$.

(a) We have

$$\begin{aligned} \int_{[0,1]} \int_{[0,1]} \frac{x^2 - y^2}{(x^2 + y^2)^2} d\lambda(y) d\lambda(x) &= \int_{[0,1]} \left[\frac{y}{x^2 + y^2} \right]_0^1 d\lambda(x) \\ &= \int_{[0,1]} \frac{1}{x^2 + 1} d\lambda(x) \\ &= \arctan(1) - \arctan(0) \\ &= \frac{\pi}{4} \end{aligned}$$

and

$$\begin{aligned} \int_{[0,1]} \int_{[0,1]} \frac{x^2 - y^2}{(x^2 + y^2)^2} d\lambda(x) d\lambda(y) &= \int_{[0,1]} \left[\frac{-x}{x^2 + y^2} \right]_0^1 d\lambda(y) \\ &= - \int_{[0,1]} \frac{1}{1 + y^2} d\lambda(y) \\ &= -(\arctan(1) - \arctan(0)) \\ &= -\frac{\pi}{4} \end{aligned}$$

- (b) It is easily checked that $f(0.25, 0.75) = -1.28$, which shows that f does not meet the nonnegativity criterion of Tonelli. On the other hand, consider a triangle with one endpoint at the origin and the other two endpoints at $(x, 0)$ and $(x, 0.5x)$. This triangle is bounded above by the line $y = 0.5x$, so we can derive the following inequality:

$$\begin{aligned} \frac{x^2 - y^2}{(x^2 + y^2)^2} &\geq \frac{x^2 - \frac{1}{4}x^2}{(x^2 + y^2)^2} \\ &\geq \frac{x^2 - \frac{1}{4}x^2}{\left(x^2 + \frac{1}{4}x^2\right)^2} \\ &= \frac{\frac{3}{4}x^2}{\frac{25}{16}x^4} \\ &= \frac{12}{25x^2}. \end{aligned}$$

Thus $\lim_{x \rightarrow 0^+} f = \infty$, so the integral over the whole space $([0, 1] \times [0, 1])$ is also ∞ , and thus f does not meet the $\mathcal{L}^1$ condition of Fubini.

□

5B.2.

- (a) Give an example of a doubly indexed collection $\{x_{m,n} : m, n \in \mathbb{Z}^+\}$ of real numbers such that

$$\sum_{m=1}^{\infty} \sum_{n=1}^{\infty} x_{m,n} = 0 \quad \text{and} \quad \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} x_{m,n} = \infty.$$

- (b) Explain why (a) violates neither Tonelli's Theorem nor Fubini's Theorem.

Proof.

- (a) Consider the collection

	$n = 1$	$n = 2$	$n = 3$	$n = 4$	$n = 5$	$n = 6$	
$m = 1$	1	0	-1	0	0	0	...
$m = 2$	0	2	0	-2	0	0	...
$m = 3$	0	0	3	0	-3	0	...
$m = 4$	0	0	0	4	0	-4	...
$m = 5$	0	0	0	0	5	0	...

Then the equalities follow as desired since the sums of the rows (i.e. $\sum_{n=1}^{\infty} x_{m,n}$ for a fixed m) are all 0, and the sums of the columns are 1 for the first column and 2 for every other column, and thus for a fixed n ,

$$\sum_{m=1}^{\infty} x_{m,n} = 1 + \sum_{m=2}^{\infty} 2 = \infty.$$

- (b) We can attempt to apply the mentioned theorems by defining $f : \mathbb{Z}^+ \times \mathbb{Z}^+ \rightarrow \mathbb{Z}^+$ by $f(m, n) = x_{m,n}$, defining μ to be counting measure on $\mathbb{Z}^+ \times \mathbb{Z}^+$, and then expressing the sums as

$$0 = \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} x_{m,n} = \int_{\mathbb{Z}^+} \int_{\mathbb{Z}^+} f(m, n) \, d\mu(n) \, d\mu(m)$$

and

$$\infty = \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} x_{m,n} = \int_{\mathbb{Z}^+} \int_{\mathbb{Z}^+} f(m, n) \, d\mu(m) \, d\mu(n).$$

Thus, switching the order of integration changes the value of the integral, and neither Tonelli's nor Fubini's Theorem holds.

This does not contradict Tonelli since f does not meet the nonnegativity requirement. It does not contradict Fubini because, noting that the product of counting measures is itself a counting measure, the integral over $\mathbb{Z}^+ \times \mathbb{Z}^+$ is ∞ , so f does not meet Fubini's criteria.

□

USES OF THEOREM 2.59 are listed in the following collaborative document:

https://docs.google.com/document/d/1VGNxxh36a41xfa0HlpcLn-g7eYjsyGJOLXUreW_Oi2c